DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025–26
HALF YEARLY EXAMINATION

CLASS: X **FULL MARKS: 80**
SUBJECT: MATHEMATICS [SET A] **TIME: 3 HOURS**

Answers to this Paper must be written on a paper provided separately.

You will not be allowed to write during the first 15minutes.

This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.

Attempt all questions from Section A and any four questions from Section B.

The intended marks for questions or parts of questions are given in brackets[ ].

This paper consists of 6 printed pages.

SECTION A

(*Attempt* ***all*** *questions from this section*)

Question1

Choose the correct answers to the questions from the given options. (Do not copy the question, Write the correct answer only.) **[15]**

(i) If nth term of an A.P. is given by (5n – 3), then the sum of its first five terms is :

(a) 30 **(b) 60** **(c) 90** **(d) 120**

(ii) The point P(2, 3) is reflected in the line x = 4 to the point P'. Find the co-ordinates of the point P'.

(a) (4, 3) **(b) (-2, 3)** **(c) (6, 3)** **(d) (3, 6)**

(iii) If A = [a b] and B = $\begin{bmatrix} c \\ d \end{bmatrix}$**, then:**

(a) Only matrix AB is possible **(b) Only matrix BA is possible**
(c) Both AB and BA are possible **(d) Both matrices AB and BA are possible, AB=BA**

(iv) The remainder when f(x) = $x^2 - 4x + 2$ **is divided by 2x + 1, is:**

(a) $\frac{1}{4}$ **(b)** $\frac{17}{4}$ **(c) - 10** **(d) 22**

(v) If ‘a’ is a natural number and one of the roots of the equation $3x^2 - 14x + 8 = 0$, then the value of ‘a’ is :

(a) 4 **(b) 8** **(c)$\frac{2}{3}$** **(d)6**

(vi) If the scale factor is 1:50000, and the distance between two cities on the map is 2cm, then the actual distance between two cities is:

(a) 100000km **(b) 2500km** **(c) 1000km** **(d) 1km**

(vii)If $25 - 4x \leq 16$, $x \in N$, then the smallest value of ‘x’ is :

(a) $\frac{9}{4}$ **(b) 2** **(c)3** **(d)4**

(viii) The value of ‘x’ in the equation $(x + 1) + (x + 4) + (x + 7) \ldots\ldots (x + 28) = 165$ is :

(a) 1 **(b) 2** **(c) 3** **(d) 4**

(ix)If p, q and r are in continued proportion then,

(a) $p : q = p : r$ **(b) $q : r = p^2 : q^2$** **(c) $p : q^2 = r : p^2$** **(d) $p : r = p^2 : q^2$**

(x) For an Intra-state sale, the CGST paid by a dealer to the Central government is ₹120. If the marked price of the article is ₹2000, the rate of GST is:

(a) 6% **(b) 10%** **(c) 12%** **(d) 16.67%**

(xi) What scale factor was used to create a 40cm tall model of a 56m rocket?

(a) 1 : 14 **(b) 1: 140** **(c) 1 : 56** **(d) 1 : 40**

(xii) Determine the slope(m) of the line : $3y - \sqrt{3}\,x = 6$

(a) $\sqrt{3}$ **(b) 1** **(c) $\frac{1}{\sqrt{3}}$** **(d) $\frac{6}{\sqrt{3}}$**

(xiii) Find the value of ‘x’ so that the line passing through (1, 5) and (x, 1) makes an angle 45^o with the positive direction of x-axis

(a) 4 **(b) -3** **(c) -1** **(d) 0**

(xiv) ₹200 shares of a company are selling at ₹80. If the company is paying a dividend of 20% then the rate of return is :

(a) 55% **(b) 50%** **(c) 60%** **(d) 25%**

(xv) Assertion (A) : The sum of the first hundred natural numbers, divisible by 5 , is 25250.
Reason (R) : The sum of first n terms of an A.P. is given by $\frac{n}{2}$ (a + l), where l is last term

(a)A is true, R is false **(b) Both A and R are true**
(c) A is false, R is true, and R is the correct reason for A
(d) Both A and R are true, and R is incorrect reason for A

Question 2

a) Aman deposited a certain sum of money every month in a recurring deposit account for 2 years. If the bank pays interest at 10% p.a. and Aman receives ₹66250 as the maturity value of the account, what sum of money did he pay every month? [4]

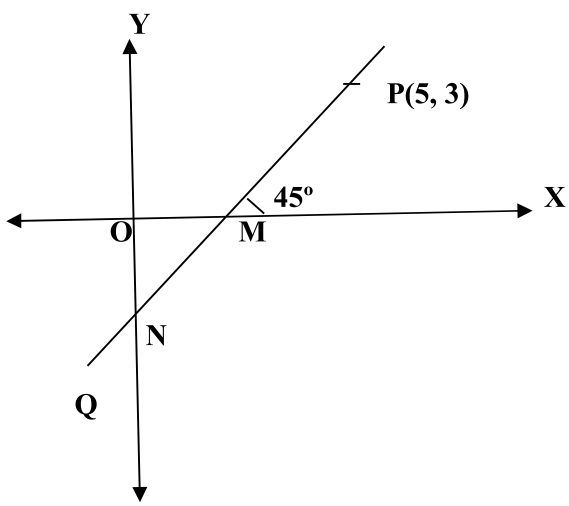
b) If $\frac{x}{b+c-a} = \frac{y}{c+a-b} = \frac{z}{a+b-c}$, using k-method, find the value of :
$(b - c)x + (c - a)y + (a - b)z$ [4]

c) Determine the number of term of the A.P. : 5, 7, 9……. So that their sum is 672 ? [4]

Question 3

a) Find the value of 'k' for which the following equation has equal roots :
$x^2 + 4kx + k^2 - k + 2 = 0$ [4]

b)



The straight line PQ meets the x axis at M and y axis at N :
(i)Write the equation of PQ.
(ii)Find the co-ordinate of M and N.
(iii)Write the equation of a line parallel to PQ and passes through origin. [4]

c) Use graph paper for this question and take 2 cm = 1 unit along both the axes.
i) Plot A(2, 2) and B(6, -2) .
ii) Reflect point A on origin to point D and write the co-ordinates of point D.
iii) Reflect A on line y = -2 to point C and write the co-ordinates of C.
iv) Find a point P on CD which is invariant under reflection in x = 0, write its co-ordinate.
iv) Write the geometrical name of the closed figure ABCD. [5]

SECTION B

(*Attempt* ***any four*** *questions from this section)*

Question 4

a) Cards numbered 2 to 101 are placed in a box. A card is selected at random from the box, find the probability that the card selected :
(i) Has a number which is a perfect square (ii) Has an odd number which is not less than 70
(iii) Has a number which is a multiple of 2 **[3]**

b) Given $\begin{bmatrix} 8 & -2 \\ 1 & 4 \end{bmatrix}$ X = $\begin{bmatrix} 12 \\ 10 \end{bmatrix}$ write down i)The order of the matrix X ii) The matrix X **[3]**

c)Find the value of 'a' and 'b' when (x – 2) and (x + 3) both are factors of $x^3 + ax^2 + bx - 12$ **[4]**

Question 5

a) Solve the inequation and represent the solution on the number line .
$\frac{3x}{5} + 2 \leq x + 4 < \frac{x}{2} + 5, x \in R.$ **[3]**

b) The 4th , 6th and the last term of a G.P. are 10, 40 and 640 respectively . If the common ratio is positive , find the first term , common ratio and the number of terms in the series.
[3]

c) Rajib has 500, ₹100 shares of a company quoted at ₹120, paying a 10% dividend. When the share price rises to ₹200 each, he sells all his shares. He invest half of the sale proceed in ₹10, 12% shares at ₹25 , and the remaining sale proceeds in ₹400, 9% shares at ₹500. Find his:

i)Sales proceed ii) Investment in ₹10, 12% shares at ₹25 iii) Original income iv) Change in income **[4]**

Question 6

a) Kavita factorized the following polynomial : $2x^3 + 3x^2 - 3x - 2$
She found the result as (x + 2)(x – 1)(x – 2).Using remainder and factor theorem , verify whether her result is correct. If incorrect, give the correct result.
[3]

b) Find the equation of the straight line perpendicular to 5x – 2y = 8 and which passes through the mid point of the line segment joining (2, 3) and (4, 5).
[3]

c) The following bill shows the GST rate, discount rate and the marked price of articles:

Item	Quantity	Rate per item (₹)	Discount	GST
A	1	₹ 10000	10%	18%
B	2	₹ 5000	5%	12%
C	1	₹ 2000	NIL	10%

Calculate i) The total amount of GST paid.
ii) The total bill amount paid. **[4]**

Question 7

a) Solve the equation $2x - \frac{1}{x} = 7$. Write your answer correct upto two decimal places. [3]

b) The line segment joining A(2, 3) and B(6, -5) is intersected by the x-axis at the point K. Write the ordinate of the K. Hence , find the ratio in which K divides AB. [3]

c) Find the value of $\frac{x+\sqrt{3}}{x-\sqrt{3}} + \frac{x+\sqrt{2}}{x-\sqrt{2}}$ by using properties of proportion, if $x = \frac{2\sqrt{6}}{\sqrt{3}+\sqrt{2}}$ [4]

Question 8

a) Each of the letters of the word AUTHORIZES is written on identical circular discs and put in a bag. They are well shuffled. If a disc is drawn at random from the bag , what is the probability that the letter is :

i)A vowel ii) One of the first 9 letters of the English alphabet which appears in a given word. iii) One of the last 9 letters of the English alphabet which appears in a given word. [3]

b) Given $\begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$ M = 6 I , where M is a matrix and I is unit matrix of order 2x2.

i)Find the matrix M [3]

c) A model of a ship is made to a scale of 1 : 200.

i) The length of the model is 6 m. Calculate the length of the ship.

ii) The area of the deck of the ship is $160000 m^2$. Find the area of the deck of the model.

iii) The volume of the model is 200 litres. Calculate the volume of the ship in m^3. [4]

Question 9

a) A line AB meets x-axis at A and y-axis at B . P(4, -1) divides AB in the ratio 1 : 2.

i)Find the coordinates of A and B

ii)Find the equation of a line passing through origin and parallel to AB. [3]

b) John invested ₹ 9600 on ₹ 100 shares at ₹20 premium paying 8% dividend. John sold the shares when the price rose to ₹160. He invested the proceed (excluding dividend) in 10% ₹50 shares at ₹40. Find the :

i) Original number of shares ii) New number of shares iii) Change in the two dividends. [3]

c) Two pipes flowing water can fill a cistern in 6 minutes. If one pipe takes 5 minutes more than the other to fill the cistern, find the time in which each pipe would fill the cistern. [4]

Question 10

a) Write down the coordinates of the image of the point (-2 , 4) under :

i)Reflection on the line x = 0

ii)Reflection on the line x = 1

iii) Reflection on the line y = 2 [3]

b) Find the 2x2 matrix which satisfies the equation .

$$\begin{bmatrix} 3 & 7 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 5 & 3 \end{bmatrix} + 2X = \begin{bmatrix} 1 & -5 \\ -4 & 6 \end{bmatrix}$$

[3]

c) The sum of the first three terms of an A.P. is 42 and the product of the first and third terms is 52. Find the first term and common difference.

[4]